

IIC_CHECK_SUM Protocol Manual

Introduction

This manual will introduce the IIC_CHECK_SUM protocol.

This protocol was originally designed as the IIC communication protocol for the HXT_Editor software, therefore, using this protocol also allows for debugging with the HXT_Editor software. The Touch Workshop platform supports debugging with the BS-eBridge tool.

IIC_CHECK_SUM (IIC_Checksum Protocol)

Protocol Introduction :

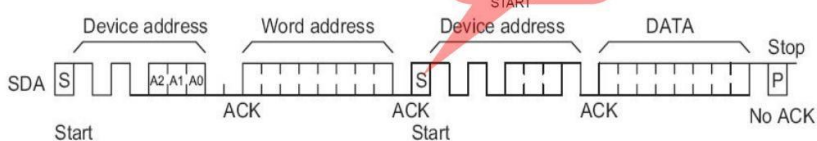
IIC_CHECK_SUM is the IIC debugging protocol, through which it is possible to not only read the environmental values, reference values, and other data of the touch buttons, but also read/write the parameters of the touch software package.

The protocol allows for the adjustment of parameters during the debugging process, enabling observation of the changes brought about by parameter modifications for better debugging of the touch system. When writing parameters using the protocol, it is necessary to perform continuous writing and send a checksum.

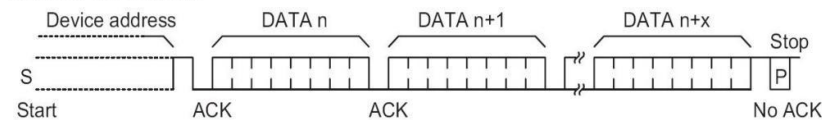
IIC Data Transmission

■ Read Timing

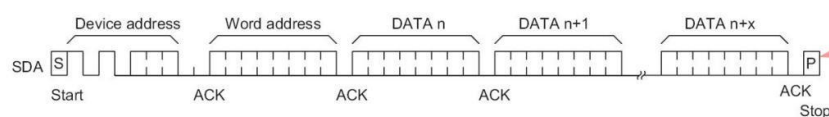
1. Read a bit of data



2. Read more data



■ Continuous data writing



1. Must be written continuously

2. When writing data, you need to check [03H].2. If [03H].2 = 1, data cannot be written. °
3. When writing data, you need to transmit a 1-byte checksum. If there is an error in the data, [03H].7 will be set to 1, and the data needs to be rewritten. °

- a. The MUC of 8KEY requires writing 17 bytes of data: 16 bytes (0xD0~0xDF) plus 1 byte checksum.
b. The MUC of 12KEY requires writing 21 bytes of data: 20 bytes (0xD0~0xE3) plus 1 byte checksum.
c. The MUC of 16KEY requires writing 25 bytes of data: 24 bytes (0xD0~0xE7) plus 1 byte checksum.
d. The MUC of 20KEY requires writing 29 bytes of data: 28 bytes (0xD0~0xEB) plus 1 byte checksum.

Data Transmission Precautions/Considerations:

- a. Transmission Speed $\leq 100K$ °
b. The way the host sends SCL/SDA signals: When sending a "high" level, set it to input mode and generate the "high" level using an external pull-up resistor. When sending a "low" level, set it to output mode and directly output the "low" level.
c. When the slave device (BS8XXXX) is busy, SCL is at a "low" level, and the host cannot transmit data during this period.
d. Before the host transmits the clock (SCL) signal, it needs to first check that SCL is at a "high" level before continuing to transmit the next bit of data.

Table 1. Address Correspondence Table

Address	Read/W rite	Purpose	Description								
00H~01H	Read	Read library version	01H=05H,00H=12H ， Version is 5.12								
02H	Read	Read the number of pressed keys	1~24 Key								
04H~07H	Reserved										
08H~0BH	Read	Read Button Status		Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
			08H	K8	K7	K6	K5	K4	K3	K2	K1
			09H	K16	K15	K14	K13	K12	K11	K10	K9
			0AH	K24	K23	K22	K21	K20	K19	K18	K17
			0BH	K32	K31	K30	K29	K28	K27	K26	K25
10H~11H	Read	KEY1_Environment Value	The environment value (ENV) is a 16-bit data, with the lower eight bits being valid.								
12H~13H	Read	KEY2_Environment Value									
14H~15H	Read	KEY3_Environment Value									
~	Read	~	The address of the environment value for KEY N follows the same pattern.								
50H~51H	Read	KEY1_Reference Value	The reference value (REF) is a 16-bit data, with the lower eight bits being valid.								
52H~53H	Read	KEY2_Reference Value									
54H~54H	Read	KEY2_Reference Value									
~	Read	~	The address of the reference value for KEY N follows the same pattern.								
B0H	Read	KEY1_RCC									
B1H	Read	KEY2_RCC									
B3H	Read	KEY3_RCC									
~			The RCC address for KEY N follows the same pattern.								
00CH~0CFH	Reserved										
0D0H~0D1H	Reserved										
0D2H	Read/Wr ite	Read/Write the Number of Pressed Keys									
0D3H	Read/Wr ite	Read/Write Shared Parameters A	Bit 7		Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
			AutoCalibrationPeriod				DebounceTimes				
0D4H	Read/Wr ite	Read/Write Shared Parameters B	MaximumKeyHoldTime					MainFreg Select		High sensitive	
0D5H	Read/Wr ite	Read/Write Shared Parameters C	MovingCalibration		PowerSave	OneKey Active		Auto Frequency Hopping			
0D6H~0D7H	Reserved										
0D8H~0DFH	Read/Wr	Read/Write Trigger Threshold Values									

	ite	for KEY1~KEY8	
0E0H~0E3H	Read/Wr ite	Read/Write Trigger Threshold Values for KEY9~KEY12	
0E4H~0E7H	Read/Wr ite	Read/Write Trigger Threshold Values for KEY13~KEY16	
0E8H~0EBH	Read/Wr ite	Read/Write Trigger Threshold Values for KEY16~KEY20	
0ECH~0EFH	Read/Wr ite	Read/Write Trigger Threshold Values for KEY21~KEY24	
0F0H~0FFH	Read/Wr ite	Reserved	